Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details Batch Number: 11

Start Date: 29th December 2025

Internship Duration: 8 Weeks

S.No	Intern Name	S.No	Intern Name
1	Marreddy Jyothi Reddy	13	Ashutosh Jha
2	Payal Shinde	14	Sowmiya P
3	R. Gowtham Mani	15	Omkar Dongare
4	Mohammad Saad Saleem Langde	16	Onkar Raut
5	Kummari Divya Sree	17	Akshita Negi
6	Gajavadan T	18	Yamini Sirisha
7	Mohammed Hamiz K	19	Kella Gayatri
8	Aman Antuley	20	Yashika Singh
9	Vennela Padiginati	21	Chetna Chaudhari
10	Karishma Namburi	22	Tiwari Shruti Santosh
11	Janhavi Sharma	23	Kodithyala Venkateshwarlu
12	Geethika Sunil	24	Jyotsna Kasibhotla

1.Project Title: ElectViz Election Data Visualization for Media- Group 2

2. Project Objective

- To integrate public election datasets and build dashboards delivering insights into vote shares, turnout rates, and party trends.

- To enable journalists and editorial teams to analyze and present complex electoral data with clarity and accuracy.
- To provide geographic mapping of election results with advanced filtering capabilities.
- To implement an AI-driven model for keyword clustering from candidate bios and manifestos.

3. Project Description in Detail

ElectViz is a Power BI-based project designed for the high-stakes environment for media requiring real-time and historical visualizations.

The platform unifies election data across multiple years at both state and national levels. By utilizing a star schema (Fact_ElectionResults, Dim_Candidate, Dim_Constituency, Dim_Party), it ensures a scalable model adaptable to future datasets.

The system visualizes vote share evolution, turnout patterns, and swing regions to enhance broadcast and digital reporting.

Key Milestones: ElectViz Project

Milestone	Description	Date Achieved
Project Kickoff	Initial briefing, team formation (25 interns), and requirement gathering for ElectViz.	29th December 2025
Data Modeling	Completion of data cleaning for 5bn+ votes and establishing Star Schema for Lok Sabha/Vidhan Sabha.	End of Week 2
Mid-Term Review	Demonstration of Historical Trends (1951-Present), Vote Share, and Seat Distribution dashboards.	End of Week 4
Feature Integration	Completion of Live Result simulations, Gender-wise distribution, and Category-wise (SC/ST/GEN) analytics.	End of Week 8
Final Submission	Final Presentation delivered, technical documentation completed, and project uploaded to GitHub.	End of Week 10

5b. Project Execution Details

The project focuses on transforming raw election data from CSVs, Excel, and APIs into structured insights using Power Query. Key performance indicators such as win/loss margins and re-election trends are tracked through dedicated modules. The final output includes broadcast-ready visuals specifically tailored for media use, allowing for deep drill-downs from regional views to booth-level insights.

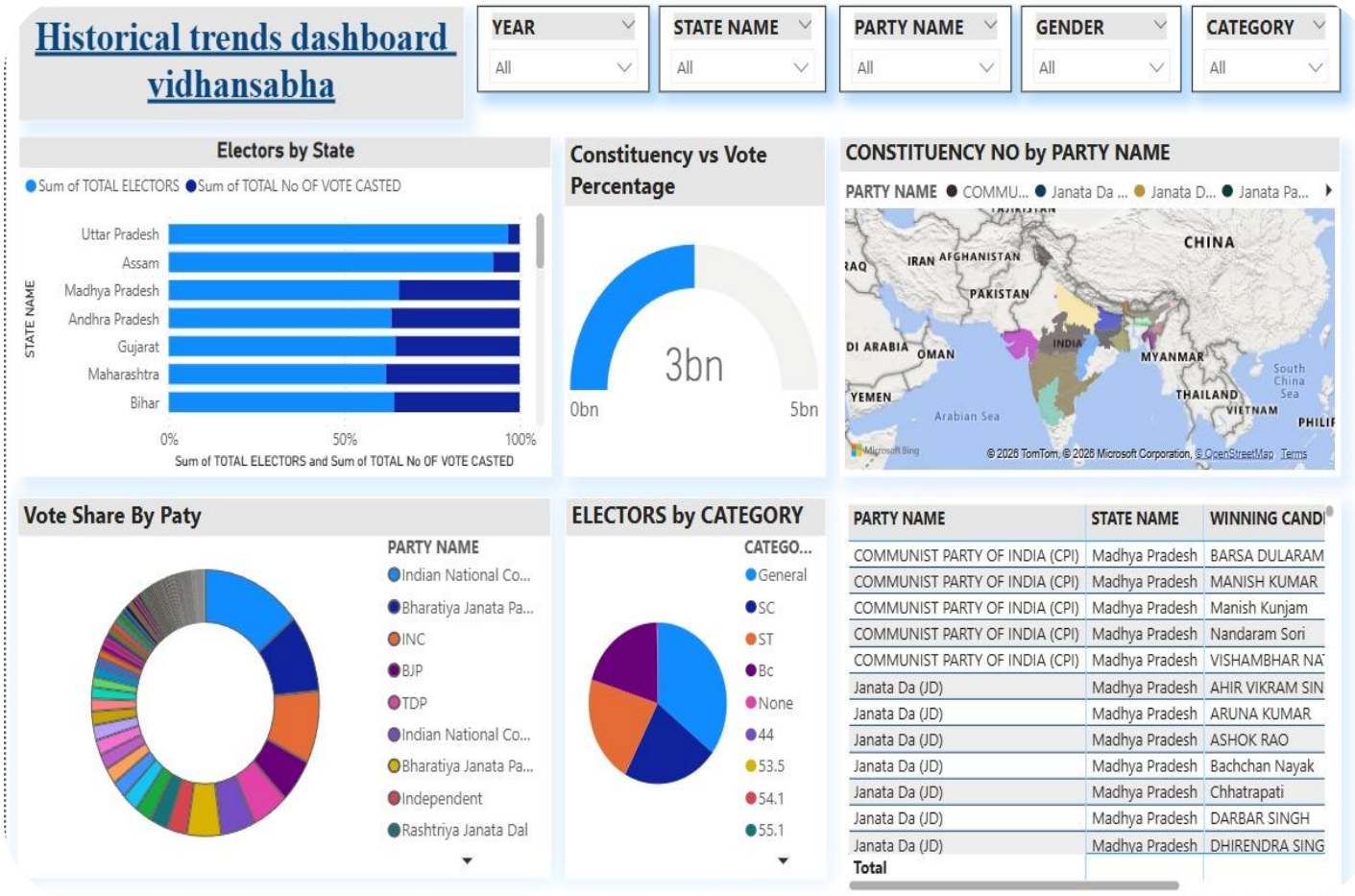
Dashboard Snapshots Overview

- **1. Historical Trends (Lok Sabha)**

  Analyzes national election patterns from 1951 to the present. It features trend lines for seat distribution over decades and yearly vote share analysis to show how party dominance has shifted over time.

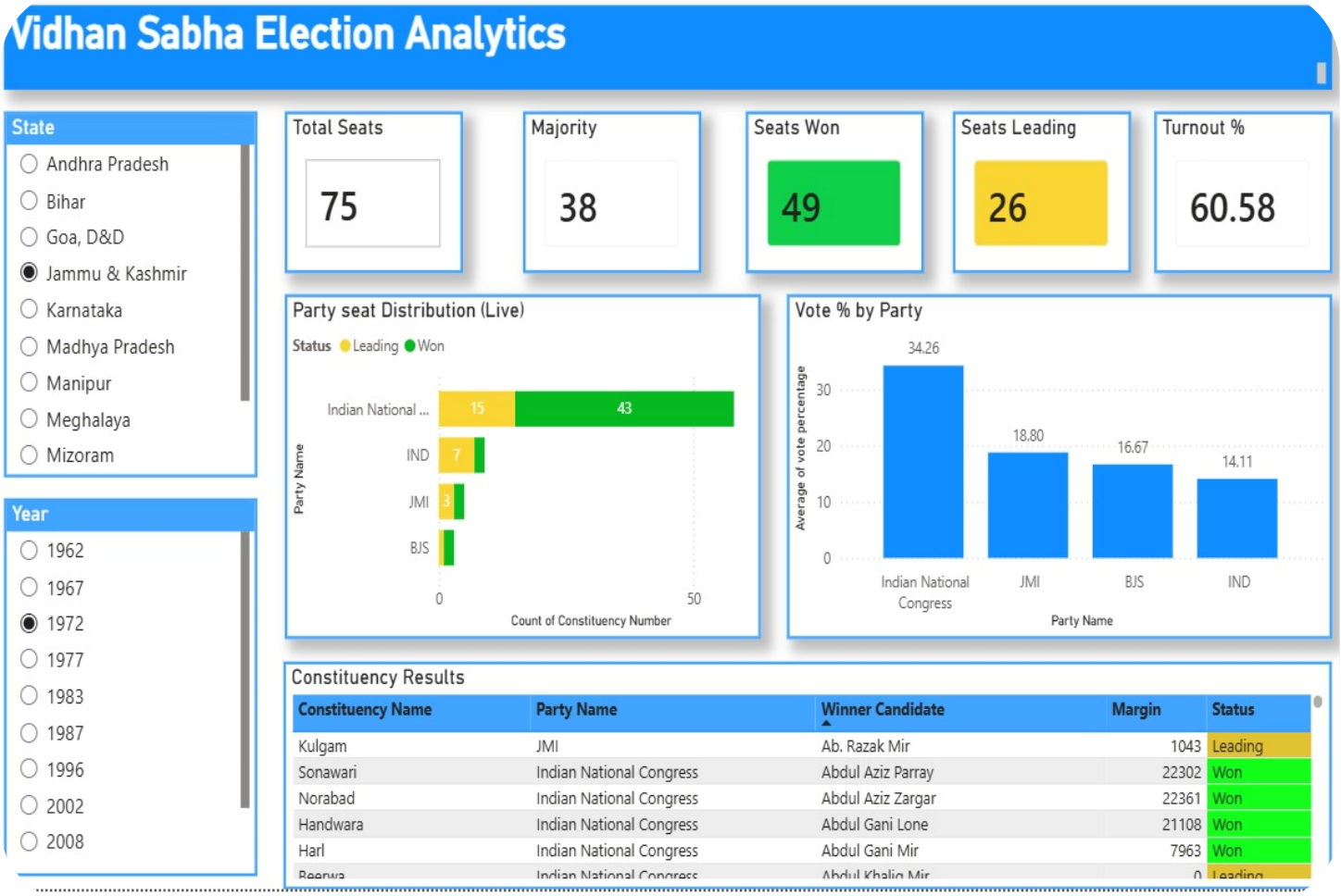
2.Historical Trends (Vidhan Sabha)

A comprehensive view of state-level elections. It includes a geographical map showing constituency distribution across India and a "Gauge Chart" to track total votes cast against the number of electors.



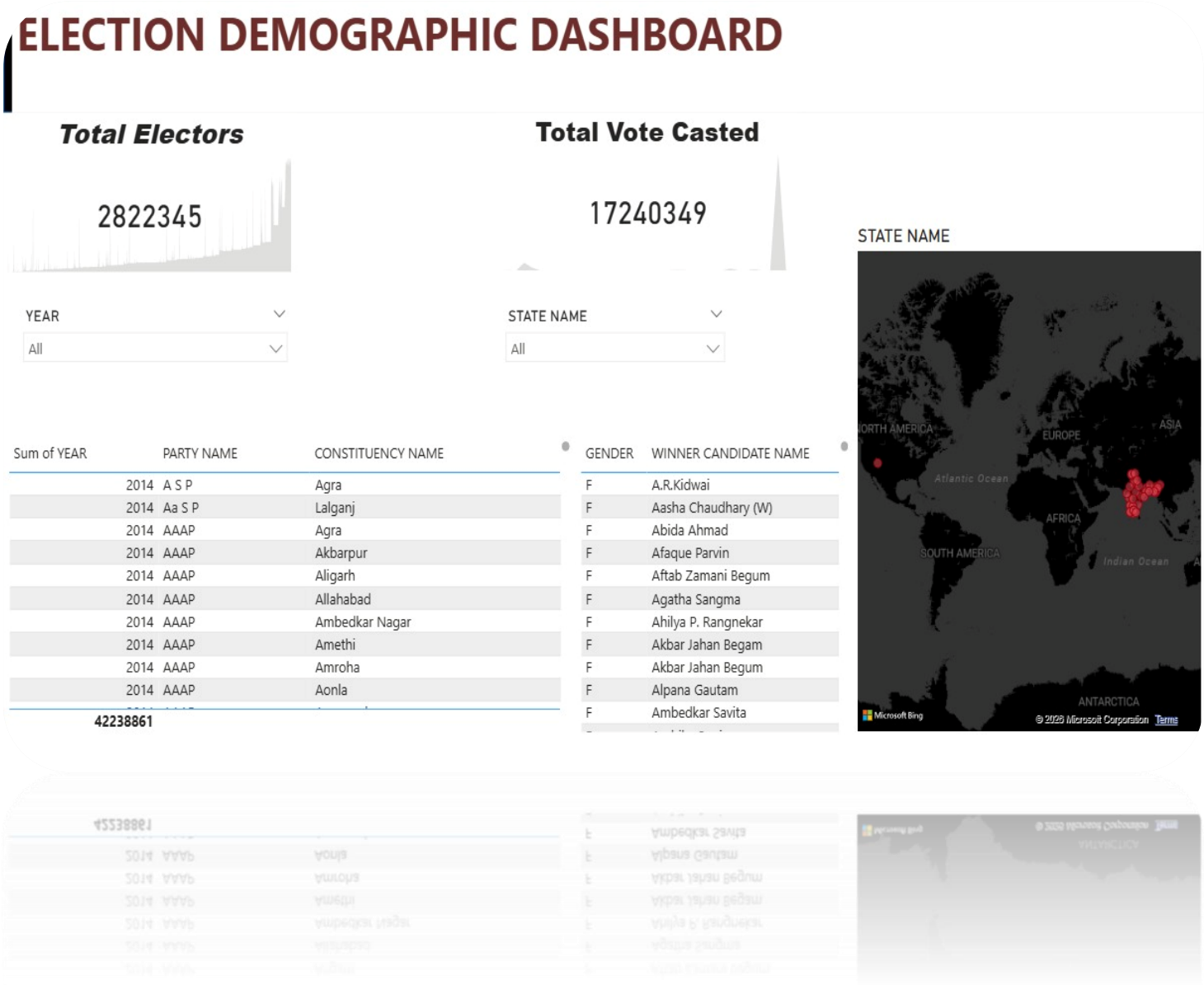
3.Live Election Analytics

A real-time simulation dashboard that tracks "Leading vs. Won" status. It uses high-visibility cards (Green for Won, Yellow for Leading) to show instant results for specific states like Jammu & Kashmir.

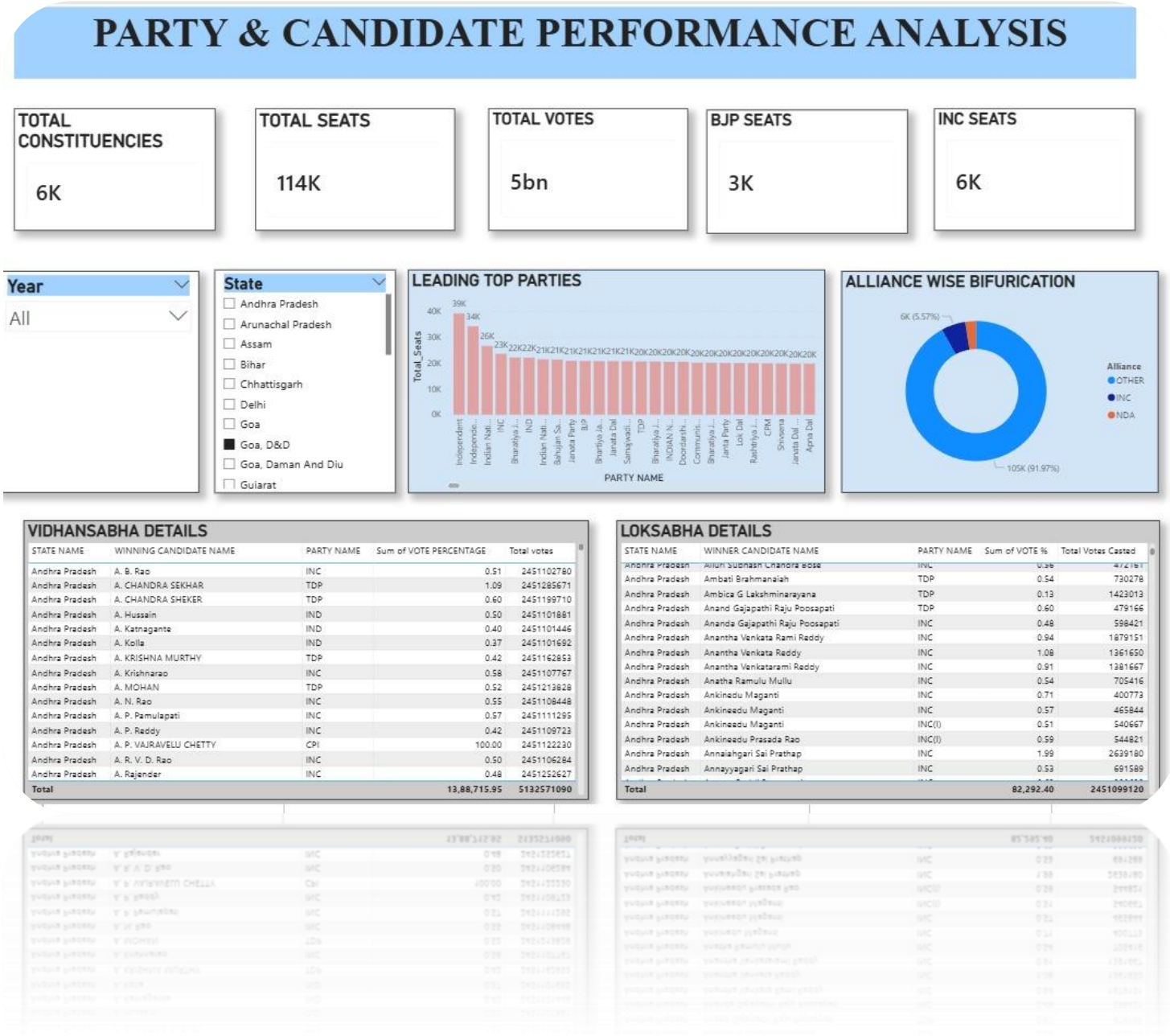


4.Election Demographic Dashboard

Focuses on the "Who" of the election. It provides a detailed breakdown of total electors versus actual votes cast, including a candidate-wise list and a global map view of data points.



1. **Party & Candidate Performance**
   A deep-dive into competitive metrics. This snapshot compares major parties (BJP, INC, etc.) and alliances (NDA vs. Others) while providing granular tables for winning candidates and their victory margins.



7. Challenges Faced

During the implementation of the **ElectViz Election Data Visualization** project, several technical and analytical hurdles were addressed to ensur broadcast-quality insights:

- **Unstructured & Noisy Data**: Election datasets from multiple years often contained inconsistencies in naming conventions for candidates and constituencies, requiring extensive cleaning and normalization.
- **Complex Data Modeling**: Designing a robust **Star Schema** with Fact_ElectionResults and multiple dimension tables was essential to maintain performance across large, historical datasets.
- **Real-Time Simulation**: Integrating live result tracking (via mock data or APIs) to mimic a high-stakes news environment required precise visual synchronization.
- **Geographic Granularity**: Ensuring accurate geographic mapping that allows users to drill down from national levels to specific constituency-level performance.

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8. Learnings & Skills Acquired

The **ElectViz** project provided a comprehensive learning pathway in data engineering and storytelling:

- **Advanced Data Visualization**: Gained expertise in using **Power BI** to create interactive dashboards, including swing analysis, turnout patterns, and party-wise performance trackers.
- **ETL & Data Modeling**: Mastered the use of **Power Query** for data ingestion (CSV, Excel, APIs) and building scalable semantic models using star schemas.
- **Electoral Analytics**: Developed a deep understanding of political data metrics, such as vote share evolution, win/loss margins, and urban vs. rural voting behavior.
- **AI Integration**: Explored AI-driven techniques like **keyword clustering** from candidate manifestos to extract qualitative insights.
- **Professional Workflow**: Learned to manage a week-wise implementation plan, meeting specific milestones from data modeling to final delivery.

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9. Conclusion

The **ElectViz** internship project successfully transformed complex, raw election data into a unified, interactive platform for a news company. By bridging the gap between historical datasets and real-time reporting needs, the team developed a scalable model adaptable to future elections.